Ch. 34 Solutions

Principles of Macroeconomics

April 14, 2026

Quick Quiz

1. (c) – A recession is a period when the economy contracts. That means production – a.k.a. real GDP – falls. When firms cut back production, they tend to lay off workers, so unemployment increases.

2. (d) – Recessions are unpredictable, so they occur irregularly.

3. (c) – According to classical macroeconomic theory, the money supply doesn't affect real variables. The GDP deflator measures the price level which is a nominal variable.

4. (a) – In the short run, the money supply affects real variables.

5. (b) – Aggregate demand and aggregate supply relate output and the price level.

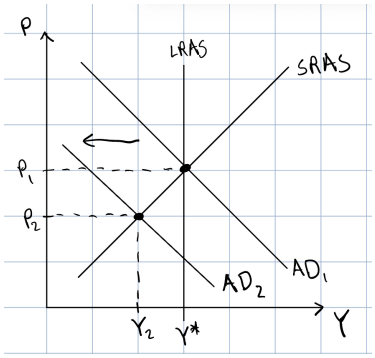
6. (b) – A fall in the price level decreases the need for people to hold money. Instead, they lend out more of their money which shifts the supply of loanable funds to the right and increases investment. An increase in investment increases aggregate demand.

7. (d) – The first two would both decrease consumption, and the third decreases government purchases. That means all three decrease aggregate demand at any given price level.

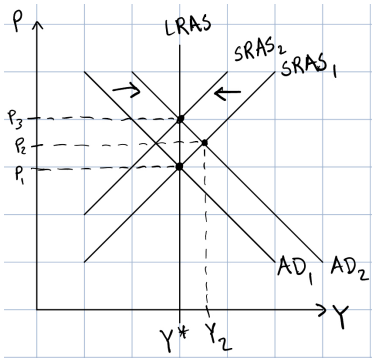
8. (d) – If wages are sticky, that means nominal wages are fixed in the short run. When the price level increases, the real value of those nominal wages decreases. If real wages decrease, then firms' costs decrease. Therefore, firms produce more. A higher price level leads to increased production. In other words, the supply curve slopes up.

9. (d) – The other three options would also shift LRAS.

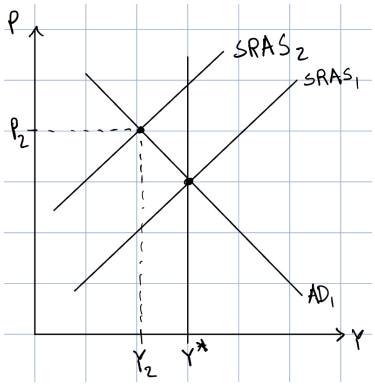
10. (c) – Business pessimism will decrease investment. That shifts AD left which leads to lower output.



11. (a) – In the short run, output increases, but in the long run, output returns to its natural level when SRAS shifts left. In the short run price level increases, but in the long run, price level increases further when SRAS shifts left.

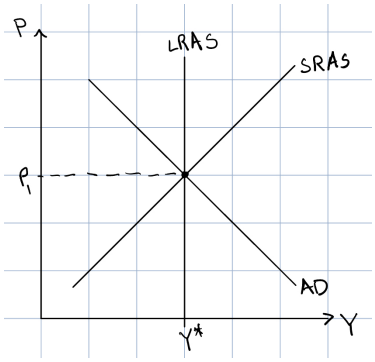


12. (c) – Stagflation is the combination of a recession and inflation. That occurs when the SRAS curve shifts left.



Questions for Review

1. Real GDP and investment decline when the economy goes into a recession. Unemployment rises when the economy goes into a recession.

2. 

3. - The wealth effect: When the price level decreases, the value of consumers' money increases, so consumers' become wealthier. In response to increased wealth, consumers consume more, so aggregate demand increases.
   - The interest-rate effect: When the price level decreases, the value of money increases, so people don't need to hold as much of it. That frees them up to lend more of their money, so the supply of loanable funds shifts to the right. As a result, investment increases and aggregate demand increases.

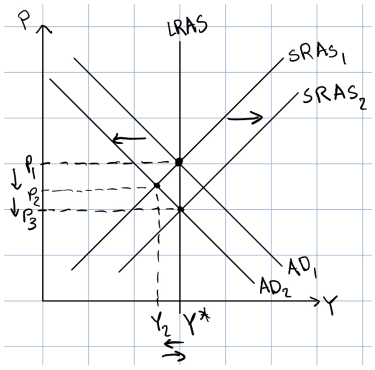
4. The LRAS curve is vertical because the price level has no effect on output in the long run. In the long run, the determinants of output are capital (physical and human), natural resources, technological knowledge, and labor. The price level doesn't affect any of those determinants in the long run.

5. - Sticky wages: Firms and workers negotiate contracts that fix workers' nominal wages in the short-run. They negotiate based on what they expect the price level to be during the contract. If the actual price level turns out to be higher than expected during the contract, then the price of firms' output increases, but the workers' wages remain constant. That means the firms make higher profits than expected which incentivizes them to increase their production. Therefore, an increase in the price level causes an increase in production.
   - Sticky prices: Firms set their prices based on what they expect the price level to be. If the actual price level turns out to be higher than expected, some firms won't immediately increase their prices because

they'll be reluctant to pay the menu costs. As a result, those firms will have lower prices than the firms that do immediately respond, so those firms will have higher sales than expected. Higher sales will incentivize the firms to increase their production. Therefore, an increase in the price level causes an increase in production.

- Misperceptions: If the price level moves higher than expected, some firms might misperceive it as an increase in the price of their particular output. That would make them think it had become more profitable to produce their good, so they would increase production. Therefore, an increase in the price level causes an increase in production.

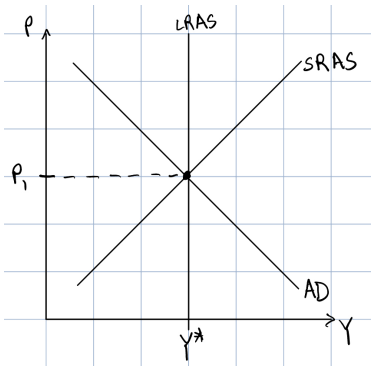
6. A stock market crash would decrease consumers' wealth. That would make them consume less and shift the AD curve to the left. In the short run, output and the price level would decrease. In the long run, SRAS would shift right, decreasing the price level further and returning the economy to its natural output level.



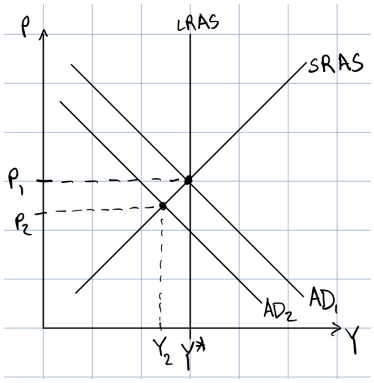
7. Extreme weather that rendered some of the capital stock temporarily unusable would shift the aggregate-supply curve to the left. In the short run, output would decrease and the price level would increase. In the long run, the recession and inflation would worsen, but eventually the SRAS curve would shift right, and the economy would return to its original long-run equilibrium.

Problems and Applications

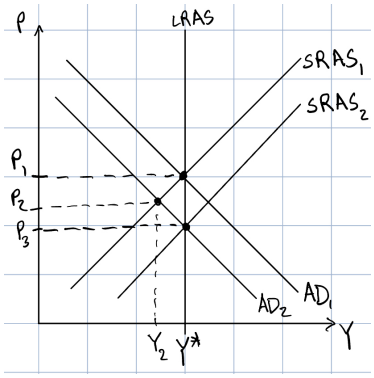
1. (a)



(b) Output and the price level fall. Because output is below it's natural level, the economy is in a recession. During a recession, unemployment increases.



(c) In the long run, firms and workers see that the price level has fallen, so they expect a lower price level than they did before the crash. Because they expect a lower price level, they will agree to lower wages than they did in the past. Firms are willing to pay less because the price of their output declines, and therefore their revenue declines. Workers will be willing to receive less because their cost of living is lower. As wages decrease, the cost of production decreases, so the aggregate supply curve shifts to the right. The economy returns to its natural output level at a lower price level.



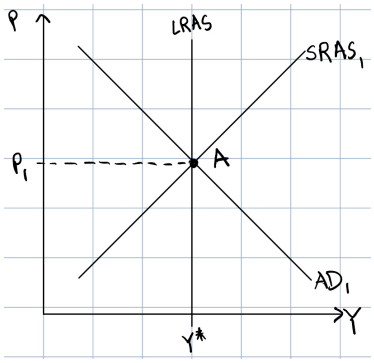
2. (a) Aggregate supply increases because the amount of labor in the economy increases.

   (b) If the labor market is perfectly competitive (an unlikely assumption), then the minimum wage will decrease employment. Less labor will decrease aggregate supply. You don't need to worry about this one.

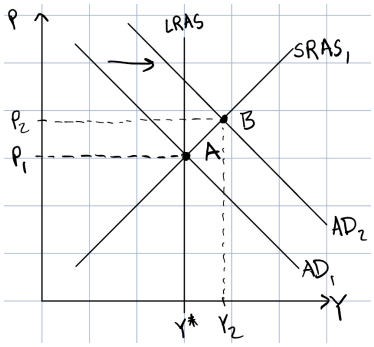
   (c) Aggregate supply increases because the country's technological knowledge increases.

   (d) Aggregate supply decreases because the capital stock decreases.

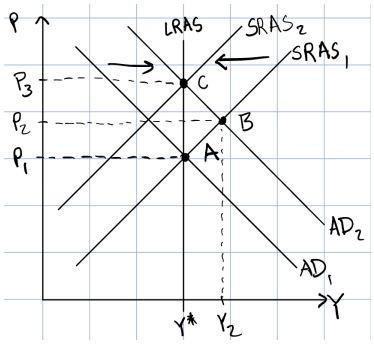
3. (a)



   (b) Output and the price level both increase.



(c) When the actual price level moves up, people begin to expect a higher price level than they did before the increase in the money supply. When the expected price level increases, the short-run aggregate supply curve shifts to the left. That moves the economy back to long-run equilibrium.

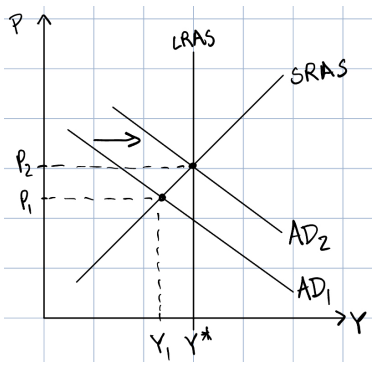


(d) In the short run, nominal wages are fixed a.k.a. "sticky," so nominal wages are the same at point A and point B. In the long run, firms and workers expect a higher price level, so they negotiate new wages that account for those expectations. Firms are willing to pay more because they expect to receive more for their output, and workers demand more because they expect higher costs of living. Therefore, nominal wages are higher at point C than point A.

(e) In the short run, nominal wages are fixed, so real wages decrease when the price level increases. Therefore, real wages decrease from A to B. In the long run, firms and workers re-negotiate wages to account for the higher price level, so real wages remain constant. Therefore, real wages are the same at A and C.

(f) It is. In the short run, an increase in the money supply increases real output and decreases real wages. In the long run, an increase in the money supply increases the price level but leaves real output and real wages unchanged.

4. If he wanted to lengthen the Christmas shopping period, he probably wanted to increase the amount of shopping households did. That means he wanted to increase consumption and shift the AD curve right. Since the economy was in a recession, a rightward shift in the AD curve would move the economy closer to natural output.



5. (a) The aggregate demand curve isn't the sum of the demand curves for different goods. That would give us the total number of goods demanded at any given price. We want the total output (a.k.a. GDP) at any given price level. Total output isn't the same as the total number of goods, and price isn't the same as price level.

   (b) Economic forces certainly affect long-run aggregate supply. Anything that changes a country's capital, natural resources, technological knowledge or labor supply will shift the LRAS curve.

   (c) Even if prices aren't sticky, wages are. If the price level increases more than expected and firms adjust their prices immediately, firms' revenues will be higher than expected, but labor costs will remain constant. That means the firms will be more profitable than expected, so they will produce more than expected. Therefore, the SRAS curve would still slope upwards.

   (d) Recessions are short-run economic fluctuations. That means the economy's short-run level of output is lower than its natural level of output. The LRAS curve remains where it is, and a shift in SRAS or AD puts the economy at an equilibrium level of output below the natural level.

6. (a) If the economy is in a recession, eventually people's expected price level will decrease. When the expected price level decreases, the SRAS curve shifts to the right for the following reasons:

   - Wages – If the expected price level falls, then workers and firms will agree to lower wages. Lower labor costs increase firms' profitability, so firms produce more, and SRAS shifts right.

- Prices – If the expected price level falls, then firms will set lower prices. Lower prices lead to increased sales, and increased sales lead to increased production. Therefore, SRAS shifts right.
- Perceptions – If the expected price level falls, then no matter the actual price level is, firms will perceive production as more profitable than they would have with their old expectation. That means they'll produce more than they would have with their old expecation, so SRAS shifts right.

(b) How quickly workers and firms adjust their expectations, wages, prices, and perceptions will determine the speed of the recovery.

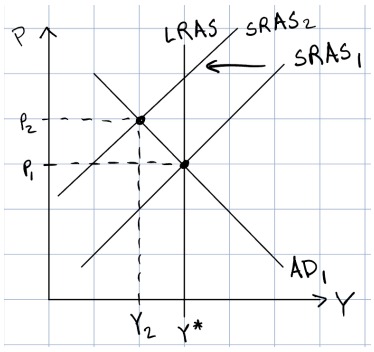
7. (a) People would expect the Fed to let inflation creep higher, so they would expect a higher price level.

(b) Workers and firms would negotiate a higher nominal wage to account for the higher expected price level. Workers would demand a higher wage because they would expect higher costs of living. Firms would be willing to pay a higher wage because they would expect higher revenues.

(c) Higher nominal wages means higher labor costs. Higher labor costs make producing less profitable at any price level.

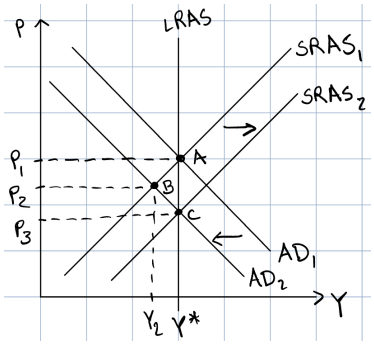
(d) A decrease in profitability would make firms produce less, so the SRAS curve would shift left.

(e) The price level would increase, and output would decrease, so we'd have stagflation.

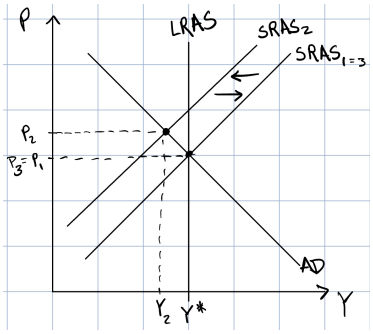


(f) Self-imposed negative supply shocks are generally not good decisions.

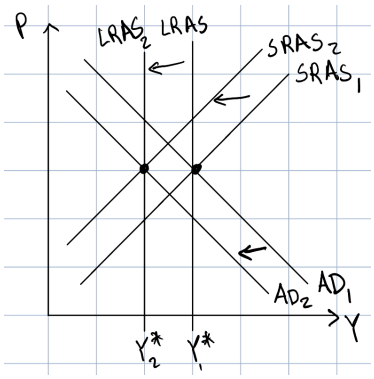
8. (a) If households decide to save a larger share of their income, consumption falls, and aggregate demand shifts left. That leads to deflation and a recession. In response to deflation, people lower their expected price level. That shifts the SRAS curve to the right which returns the economy to its natural level of output and decreases the price level even further.



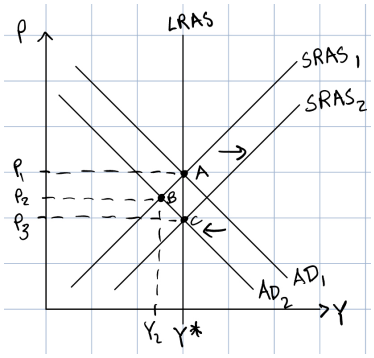
(b) If Florida orange groves suffer a prolonged period of below-freezing temperatures, the country experiences an adverse shock to its natural resources, and SRAS shifts left. That decreases total output and increases the price level. In the long run, the temperatures will return to normal, or long-run aggregate supply will shift left as well. If temperatures return to normal, the SRAS curve shifts right and returns the economy to its original price level and natural level of output (pictured below). If long-run aggregate supply shifts left, the price level becomes permanently higher and the natural level of output falls.



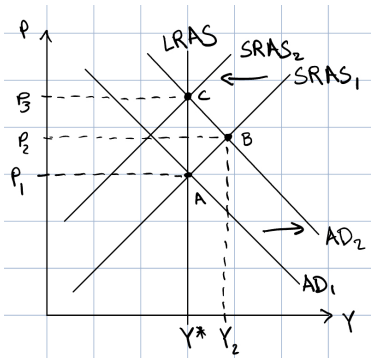
(c) If many people emigrate, the number of workers and the number of consumers will decrease. That will shift AD, SRAS, and LRAS, to the left. Output will unambiguously decrease in the short and long run. The change in the price level is ambiguous in the short and long run. The change in the price level depends on the relative size of the shifts in AD and AS.



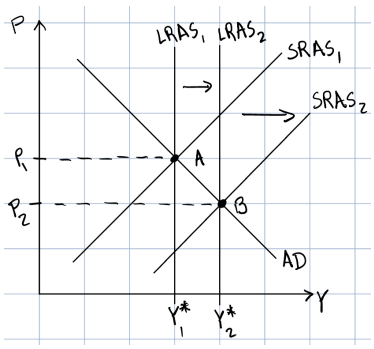
9. (a) If the stock market declines sharply, reducing consumers' wealth, consumers will consume less. That shifts AD to the left which decreases the price level and output. In response to the falling price level, people decrease their expected price level. That shifts the SRAS curve to the right, decreases the price level further, and increases output back to its natural level.



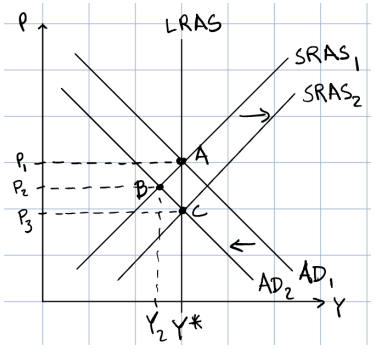
(b) If the federal government increases spending on national defense, government purchases increase and aggregate demand shifts right. That leads to increased output and inflation. In response to inflation, people increase their expected price level. That shifts the SRAS curve to the left which returns the economy to its natural level of output and increases the price level even further.



(c) A technological improvement shifts the LRAS and SRAS curves to the right. That lowers the price level and increases the level of output in the short run and the long run.

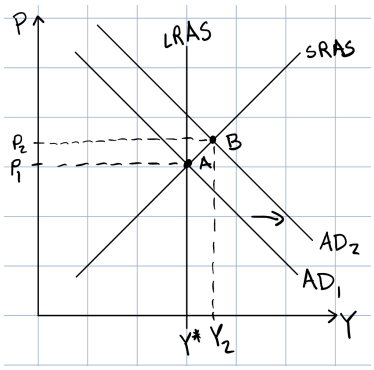


(d) If foreigners buy fewer U.S. goods, then net exports fall, and aggregate demand shifts left. That decreases output and the price level. In response to the deflation, people lower their expected price level. That shifts the SRAS curve to the right which returns the economy to its natural level of output and decreases the price level even further.

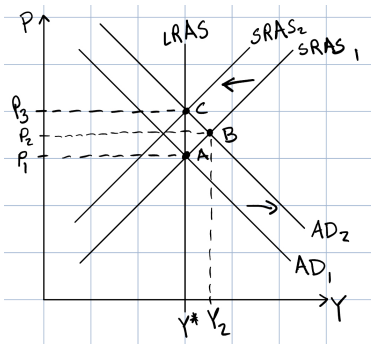


10. (a) If firms invest heavily in new capital, then investment increases, and aggregate demand shifts to the right. When aggregate demand ini-

tially shifts right, the price level is P_1. At that price level, aggregate demand is greater than aggregate supply. In order to secure some of the scarce output, people start to bid up the prices of goods. That increases the price level. As a result, firm revenues increase. In the short run, workers' wages stay constant because wages are fixed in the short run. Since revenues increase and labor costs stay constant, production becomes more profitable for firms. Firms respond by producing more, and the aggregate quantity of output supplied increases.



(b) In part (a), the price level increased. When the price level rises, people increase their expected price level. That shifts the SRAS curve to the left. Immediately after the shift, the price level is still P_2. At that price level, aggregate demand is greater than aggregate supply, so people bid up the price level in order to secure some of the scarce output. As a result, the price level rises to P_3.



When the price level rises, it has two effects on aggregate demand. First, it decreases the value of people's money which decreases their real wealth and makes them less likely to consume. Second, it increases people's desire to hold money because they have higher costs. That decreases the supply of loanable funds which increases the interest rate and decreases investment. As a result, consumption falls,

investment falls, and the aggregate quantity of output demanded falls.

(c) In the long-run, an investment boom could increase the country's capital stock. An increase in the capital stock will shift the long-run aggregate-supply curve to the right.